# CARDIO-ALPHA: Phase III Trial of Cardivex in Stage 1-2 Hypertension

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### *Background*

Cardivex is a fictional long-acting angiotensin-converting enzyme (ACE) inhibitor. CARDIO-ALPHA evaluated its antihypertensive efficacy and tolerability against placebo over 12 weeks.

### *Cohort and methods*

A total of 842 adults (mean age 58.4 years, 46% female) with stage 1-2 essential hypertension were randomized 1:1 to Cardivex 10 mg (n=421) or placebo (n=421) at 37 fictional sites. The primary endpoint was the change in mean seated systolic blood pressure (SBP) from baseline to week 12.

### *Results*

Mean SBP decreased by 14.2 mmHg with Cardivex versus 6.1 mmHg with placebo (difference -8.1 mmHg; 95% CI -9.6 to -6.6; p<0.001). Diastolic blood pressure decreased by 8.3 mmHg versus 3.2 mmHg. Blood pressure control (<140/90 mmHg) at week 12 was achieved in 61.5% versus 33.8% of patients.

### *Safety*

The most common adverse event was dry cough (7.3% vs 1.9%). Hyperkalemia (serum potassium >5.5 mmol/L) occurred in 1.8% vs 0.5%. Two cases of angioedema (0.5%) were reported with Cardivex and none with placebo. Discontinuation due to adverse events was 3.1% vs 1.4%.

### *Conclusion*

Cardivex 10 mg produced a clinically meaningful SBP reduction with an adverse-event profile typical of the ACE inhibitor class.

### *Summary table*

| Endpoint (week 12) | Cardivex (n=421) | Placebo (n=421) |
|---|---|---|
| Change in SBP, mmHg | -14.2 | -6.1 |
| Change in DBP, mmHg | -8.3 | -3.2 |
| BP control <140/90 | 61.5% | 33.8% |
| Dry cough | 7.3% | 1.9% |
| Hyperkalemia | 1.8% | 0.5% |